

The parameters of the porcine eyeball

Irene Sanchez · Raul Martín · Fernando Ussa ·
Ivan Fernandez-Bueno

Received: 15 June 2010 / Revised: 4 January 2011 / Accepted: 5 January 2011 / Published online: 2 February 2011
© Springer-Verlag 2011

Abstract

Background The eye of the domestic pig (*Sus scrofa domestica*) is an ex vivo animal model often used in vision sciences research (retina studies, glaucoma, cataracts, etc.). However, only a few papers have compiled pig eye anatomical descriptions. The purpose of this paper is to describe pig and human eye anatomical parameters to help investigators in their choice of animal model depending on their study objective.

Methods A wide search of current medical literature was performed (English language) using PubMed. Anteroposterior axial length and corneal radius, astigmatism, vertical and horizontal diameter, and pachymetry (slit-scan and ultrasound) were measured in five enucleated pig eyes of animals 6 to 8 months old.

Results Horizontal corneal diameter was 14.31 ± 0.25 mm (CI 95% 14.03 mm–14.59 mm), vertical diameter was

12.00 ± 0 mm, anteroposterior length was 23.9 ± 0.08 mm (CI 95% 23.01 mm–29.99 mm), central corneal ultrasound pachymetry was 877.6 ± 13.58 μ m (CI 95% 865.70 μ m–889.50 μ m) and slit-scan pachymetry was 906.2 ± 15.30 μ m (CI 95% 892.78 μ m–919.61 μ m). Automatic keratometry (main meridians) was 41.19 ± 1.76 D and 38.83 ± 2.89 D (CI 95% 40.53D–41.81D and 37.76D–39.89D respectively) with an astigmatism of 2.36 ± 1.70 D (CI 95% 1.72D–3.00D), and manual keratometry was 41.05 ± 0.54 D and 39.30 ± 1.15 D (CI 95% 40.57D–41.52D and 38.29D–40.31D respectively) with an astigmatism of 1.75 ± 1.31 D (CI 95% 0.60D–2.90D). **Conclusion** This paper describes the anatomy of the pig eyeball for easy use and interpretation by researchers who are considering their choice of animal model in vision sciences research.

The authors have no proprietary, financial or commercial interest in any material or method mentioned in this study.

I. Sanchez (✉)
IOBA-Eye Institute and CIBER-BBN, University of Valladolid,
Valladolid, Spain
e-mail: isanchezp@ioba.med.uva.es

R. Martín
IOBA-Eye Institute and Department of Physics
TAO – School of Optometry,
University of Valladolid,
Valladolid, Spain

F. Ussa
IOBA-Eye Institute, Ophthalmologist, Glaucoma Unit,
University of Valladolid,
Valladolid, Spain

I. Fernandez-Bueno
IOBA-Eye Institute, University of Valladolid,
Paseo de Belén nº 17 Campus Miguel Delibes,
47011 Valladolid, Spain

Keywords Porcine cornea · Porcine parameters ·
Porcine anatomy · Porcine eye · Pig eye · Pig cornea ·
Pig parameters · Pig anatomy

Introduction

The pig eye is an ex vivo animal model often used in vision sciences research because its morphology is similar to the human eye [1–6]. The pig eye has been used in neuroretinal studies due to the similarity of the distribution of the retinal layers to that of the human retina [1], and it is also a validated animal model of glaucoma [4]. Moreover, it has been used in cataract surgery research [5], in corneal transplant studies [6, 7], and as a model in aberrometry studies [2, 3].

A few studies compile a complete description of the pig eye [8]; some eye parameters can be found in papers [3, 4] that provide information about the similarities and

differences between pig and human eyes, but only as they pertain to the study. Moreover, there are no papers that collect anatomical characteristics about corneal curvature and refractive power measured by non-invasive techniques (autokeratometer or corneal topography).

The purpose of this paper is to summarise the anatomical parameters of the pig eye, including corneal topography description, to help vision researchers choose an animal model that is appropriate to the objectives of their study.

Materials and methods

This paper presents two principal sections. First, a PubMed search of the English language literature was performed looking for human eye [9, 10] and pig eye parameters. Second, an experimental study to determine the anatomical parameters of the enucleated pig eye was completed.

For the experimental study, five enucleated eyes from five pigs (*Sus scrofa domestica*) were obtained from the local abattoir. Animals were white (not albino) domestic pigs between 6 and 8 months of age, and they weighed 120–150 kg. The eyeballs were enucleated around 8:30 a.m. and transported to the laboratory on ice. The measurements were made between 9:00 a.m. and 12:00 a.m. The excess of tissue external to the eyeballs, including muscles, the lacrimal gland, and conjunctiva, was removed with 0.12 mm Castro titanium delicate forceps and Wescott scissors (John Weiss International, Milton Keynes, UK). Pig eyes were always kept in DMEM culture medium (Dulbecco's Modified Eagle's Medium) supplemented with an antibiotic/antimycotic mixture (Gibco, UK). The eyeballs were cannulated for maintenance of the intraocular pressure within normal values (15 mmHg) measured with Perkins tonometry. Anteroposterior eye length, corneal radius, astigmatism, vertical and horizontal diameter, and pachymetry (slit-scan and ultrasound) were measured.

Corneal curvature was measured with a portable autokeratometer (ARK-30 Nidek, Fremont, CA, USA) [11] and with a manual keratometer (OM-4 Topcon, Tokyo, Japan). Central corneal thickness was measured with ultrasound pachymetry (Sonogage Corneo-Gage Plus, Renaissance Parkway, Cleveland, OH, USA) calibrated by the manufacturer. Finally, a topographic study of the corneal surface was conducted with an ORBSCAN II (version 3.12: Bausch & Lomb, Rochester, NY, USA) determining central and peripheral corneal thickness, corneal curvature, refractive power, and corneal diameter.

Statistical analysis

The average value with standard deviation (SD) and confidence interval at 95% (CI 95%) of each parameter measured were determined with SPSS 14.0 for Windows.

Results

Literature search results

Sclera

The pig sclera is negatively charged, probably due to the presence of sulphate and uronic groups in glycosaminoglycans at pH 7.4, and the isoelectric point is three, which is the same as that of the human sclera [12]. The scleral spur has a special feature: there is a small, noticeable structure in the nasal quadrant. There are variations in the scleral sulcus depth that depend on the location in the circumference of the limbus. In the temporal quadrant, the scleral spur is characteristic. In the inferior and nasal quadrants, the scleral sulcus is obvious, whereas in the superior quadrant it is not easily visible [13]. There is little nerve innervation in this area [14]. The thickness of the scleral wall ranges from 830 to 1250 μm [12, 15], and the anterior segment is strongly pigmented [16]. The water content is $69.5 \pm 1.18\%$ [15]. Histologically, pig sclera is very similar to human sclera, although pig collagen appears more disorganised than human collagen [15].

Cornea

The horizontal corneal diameter of the pig eye is 14.23 mm, and the vertical diameter is 12.09 mm [8]. The ultrasound pachymetry is between $1013 \pm 10 \mu\text{m}$ in an ex vivo model [17] and 666 μm (with a range of 534 μm to 797 μm) in a live animal [18]. The corneal epithelium has a thickness of 80 μm ; this thickness is not constant, and can vary by 25 μm . The stromal thickness is 900 μm , and Descemet's membrane with the endothelium has a thickness of approximately 30 μm . Bowman's layer is absent in porcine cornea [17]. The stroma has a large amount of collagen type I [17], with mainly circumferential orientation [19]. Descemet's membrane extends beyond the origin of the cornea, and inserts slightly in the short pectinate ligaments [13].

The water content of the porcine cornea is $71.93 \pm 0.47\%$, and shows a transparency of $54.77 \pm 0.47\%$ [20].

No papers were found with a description of pig eye corneal topography curvature obtained with computerised corneal topographies.

Lens

The lens is composed mainly of three types of proteins called crystallins that can be soluble or insoluble. The soluble proteins are α -, β - and γ -crystallins. The α -crystallins make up 35% of the outer lens and 22% of the inner lens (nucleus). The β -crystallins make up 45% of the outer lens and 35% of the inner lens. The γ -crystallins are present in smaller amounts than α -crystallins and β -crystallins, and are found in a greater

proportion in the lens nucleus than in the cortex. The increase of γ -crystallins in the inner lens may contribute to the refractive index gradient [21].

The insoluble proteins of the lens have a higher concentration in the cortex than the nucleus, with an estimated concentration of 25% [21].

The refractive power of the lens is 49.9 ± 1.5 D, with a refractive index of 1.4686 and a negative spherical aberration of -3.6 ± 2.0 D. Its anterior and posterior radii are 7.08 ± 0.35 mm and 5.08 ± 0.14 mm respectively. The lens thickness is 7.4 ± 0.1 mm [3].

The minimum vaulting to introduce an intraocular lens (ICL) in a minipig eye (with 800 μ m of corneal thickness [22]) is at least 150 μ m, which corresponds to 1/3 to 1/4 of the total thickness of the cornea. The anterior chamber depth is approximately 3.5 mm [23].

Ciliary body

The ciliary body is a structure with abundant vascularisation and innervation. The stroma contains melanocytes with a double-layer epithelium and a pigmented and a non-pigmented layer. In general, the pig eye ciliary body is more pigmented than the human eye ciliary body [16]. In the nasal zone of the ciliary body, there are a few fibres of the ciliary muscle radially oriented. In the temporal zone, closest to the iris, there is a simple organisation of muscle fibres circumferentially distinct from radial fibres. In this zone, fibres are longer and reach the sclera and the scleral spur, although other fibres reach the back of the trabecular meshwork [13]. The cells of the ciliary muscle, myofilaments, do not show disciform or parallel organisation. The density of bands in the cell surface is similar to smooth muscle cells of the blood vessels and the wall of the human bowel. A basal membrane and connective tissue make up an elastic network that surrounds these cells. Cross-sections of the ciliary body show a fine network of radial fibres that extend from the choroid through the ciliary muscle to the trabecular meshwork [14].

McMenamin et al. [13] demonstrated that the muscle fibres are longer in the superior and inferior quadrants, the ligaments are more robust in the nasal and temporal quadrants, and the pigmentation is not homogeneous throughout the whole circumference.

Uveoscleral drainage

The iridocorneal angle appears to be a heterogeneous measure between animals, as shown by Bartholomew [8]. Uveoscleral drainage was measured to evaluate choroid drainage. The intraocular pressure is constant at 10 mmHg, with an average drainage of 2.8 ± 0.9 μ l/min. After blocking the conventional pathway of drainage, the drainage declined at an average rate of 1.1 ± 0.5 μ l/min. However,

blocking the vortex veins does not change the uveoscleral drainage, which is maintained at 1.2 ± 0.8 μ l/min [24].

The postoperative inflammatory response of the pig eye is greater than that of the human eye; thus, posterior pole surgery may trigger diffuse choroidal haemorrhage that sometimes may be unstoppable [25]. The choroidal blood flow, with the retinal artery clamped, is 500 μ l/min [26].

Lamina cribrosa

The structure of the lamina cribrosa has been studied by second harmonic generation imaging using a scanning laser ophthalmoscope-based microscope. With this technique, the collagen fibres that form the lamina cribrosa and their dehydration have been observed [27].

Retina

Pig retina maintains the structure of ten layers, the same as in the human retina because embryonic development is similar [28]. The outer plexiform layer [29] contains the synapses between the photoreceptors and second-order neurons, bipolar and horizontal cells. The inner plexiform layer is the other synaptic zone and connects third-order neurons, like amacrine and ganglion cells, with the bipolar cells. Müller cells are the main retinal glial cells. Müller cells are the main retinal glial cells. They extend through most of the retina from the outer segments, where their finger-like processes form the outer limiting membrane to their basement membrane, which makes up the inner limiting membrane [29]. The attachment between the retinal pigment epithelium and Bruch's membrane is mediated by the interaction between integrins (the main components of hemidesmosomes) on the retinal pigment epithelium surface and ligands in the extracellular matrix. The hemidesmosomes present in the basal surface of the retinal pigment epithelium maintain cohesion between the epithelium and Bruch's membrane [30].

There is a depressed central area, rich in cones, that is comparable to the human macula [25, 28]. It has the shape of a horizontal band called the foveal streak [31], which is placed over the optic nerve head [16]. The density of cones in this area is between 15,000 and 40,000 cells/mm² [16].

The proportion of cones and rods is similar to the human retina, and the paramacular density of cones is also similar [25]. These data are obtained thorough multifocal electroretinography. The fundus reflection is orange to pale grey, with pigmented epithelial cells. The pig eye lacks a tapetum [32]. The retinal circulation is holangiotic [25].

Vitreous

The vitreous is a gel containing collagen and sodium hyaluronate. Sodium hyaluronate has a coil-shaped

molecular structure, and is uniformly distributed in a three-dimensional network of collagen fibres that form a triple helix structure. The volume of the polymer network of the two molecules is only 1–2% of the total volume; the other 98–99% is composed of water. Moreover, the vitreous body represents 80% of the total volume of the eyeball. Its functions are eyeball form maintenance, mechanical stress absorption, maintenance of the homoeostasis of the eye and adjustment of the position of the lens [33]. In addition, the collective diffusion coefficient of the vitreous body is similar to the aqueous humour [29], with a calculated elastic modulus of 57.3 ± 5.5 Pascals [34].

Innervation of the eyeball

The temporal quadrant of the posterior pole is characterised by the entry of the long ciliary nerves [14]. Glutamate is the major neurotransmitter in the neuronal cells of the visual pathway, and the presence of a GoG protein, which is expressed in the metabotropic glutamate receptors of bipolar cells of mammals, is observed [35].

Vascularization of the eyeball

The pig eyeball receives most of its blood supply through the long and short posterior ciliary arteries and the chorioretinal artery. The ciliary processes are fed by the iridociliary artery that forms a ring, the origins of which are the long posterior ciliary arteries [36].

The iris receives its blood supply from the iridociliary arterial circle, the origins of which are the anterior ciliary arteries. Capillaries are oriented radially in the direction of the pupil, and the morphology of the capillaries is characterised by a spiral or zig-zag. The veins also have an undulating morphology and drain into the vortex veins through pars plana veins [36].

Extrinsic ocular motility

The sixth extraocular muscles of the pig are similar to those of the human eye; moreover, the pig eye has a seventh extra ocular muscle that surrounds the optic nerve, and blood vessels called the retractor bulbi muscle. This muscle tends to retract the eyeball into the orbit [25].

Eyelids

The pig eye has a nictitating membrane [25]. The nictitating membrane is located in the medial angle of the two eyelids. The third eyelid mechanically protects the cornea, spreads the tear film and offers local immunological defences provided by the substances produced in lymph nodules. The superficial gland of the third eyelid produces part of the tear film [37]. The movement of the third eyelid is passive and depends on the action of the retractor bulbi muscle: the retraction of the eyeball and third eyelid protrusion towards the temporal angle [37]. The third eyelid cartilage consists of the dorsal and ventral branches and a crossbar resembling an anchor [37]. The deep gland of the third eyelid is also called the Harderian gland. It has a lobular structure, and is situated inside the periorbit on the medial orbit wall. The secretion of this gland is a lubricant that covers the eyeball and has antibacterial and immunological properties [38].

Experimental results

Table 1 summarises the results (average, standard deviation and 95% confidence interval) of the experimental measurements of enucleated pig eye parameters. The differences between porcine anterior segment and human anterior segment are shown in Table 2. Figure 1 shows corneal topography representative of one of the examined pig eyes. Figure 2 shows the topography of a healthy human cornea, to graphically show the differences between pig and human corneas.

Table 1 Data obtained from experimental measurements. Average with standard deviation (SD) and the confidence interval at 95% (CI 95%) was calculated

Ø=diameter; K=keratometry; R1=steeper main corneal meridian; R2=flatter main corneal meridian; ^a data obtained with ARK-30 automated keratometer; ^b data obtained with OM-4 manual keratometer. Corneal astigmatism was calculated as the difference between the power of the main corneal meridians (R1 and R2)

Parameter measured	Average $\pm$ SD	Confidence interval 95%
Ø Horizontal visible iris	14.31 $\pm$ 0.25 mm	14.03–14.59 mm
Ø Vertical visible iris	12 $\pm$ 0.0 mm	–
Ø Anteroposterior	23.9 $\pm$ 0.08 mm	23.01–29.99 mm
Ultrasonic pachymetry	877.6 $\pm$ 13.58 μ m	865.70–889.50 μ m
Slit-scan pachymetry	906.2 $\pm$ 15.30 μ m	892.78–919.61 μ m
K automatic in R1 ^a	41.19 $\pm$ 1.76 D	40.53–41.86 D
K automatic in R2 ^a	38.83 $\pm$ 2.89 D	37.76–39.89 D
Corneal astigmatism ^a	2.36 $\pm$ 1.70 D	1.72–3.00 D
K manual in R1 ^b	41.05 $\pm$ 0.54 D	40.57–41.52 D
K manual in R2 ^b	39.30 $\pm$ 1.15 D	38.29–40.31 D
Corneal astigmatism ^b	1.75 $\pm$ 1.31 D	0.60–2.90 D

Table 2 Comparison of average parameters measured in the pig eyeball and estimated average values of the human population according to the literature

Parameter measured	Pig eye	Human eye
Ø Horizontal visible iris	14.31 mm	11.7 mm
Ø Vertical visible iris	12 mm	10.6 mm
Ø Anteroposterior	23.9 mm	24 mm
Ultrasonic pachymetry	877.6 µm	520 µm
Mean corneal radius	8.45 mm	7.80 mm

Discussion

Sclera

Pigmentation of the anterior sclera made regular trans-scleral infrared fundus illumination impossible. This illumination is feasible in human eyes [16].

Some papers show that the human and porcine sclera have comparable permeabilities, and thus pig sclera is an excellent model for studying the pharmacokinetics of trans-scleral drug delivery in vivo [39]. However, Nicoli

et al. [15] showed that the different thicknesses of the human and porcine sclera have to be taken into account, even though the permeability of porcine sclera is the same as the human.

Cornea

Porcine corneal thickness is almost twice that of the human cornea, and lacks Bowman's layer [17]. Although Bartholomew [8] showed that pig cornea have Bowman's layer, later histological studies [17] did not find Bowman's layer. The thickness of the cornea as measured with ultrasound pachymetry is 877 ± 14 µm (CI 95% 866–890 µm), which is less than the value obtained by Jay [17] (1013 ± 10 µm) and higher than that of Faber [18] (666 µm). This difference could be due to the different ages of the animals slaughtered or the type of pig used. Additionally, in ex vivo models, corneal pachymetry could increase due to corneal swelling if measurements are taken a long time after animal sacrifice.

There are no previous papers describing the porcine corneal topography, especially the radius of corneal

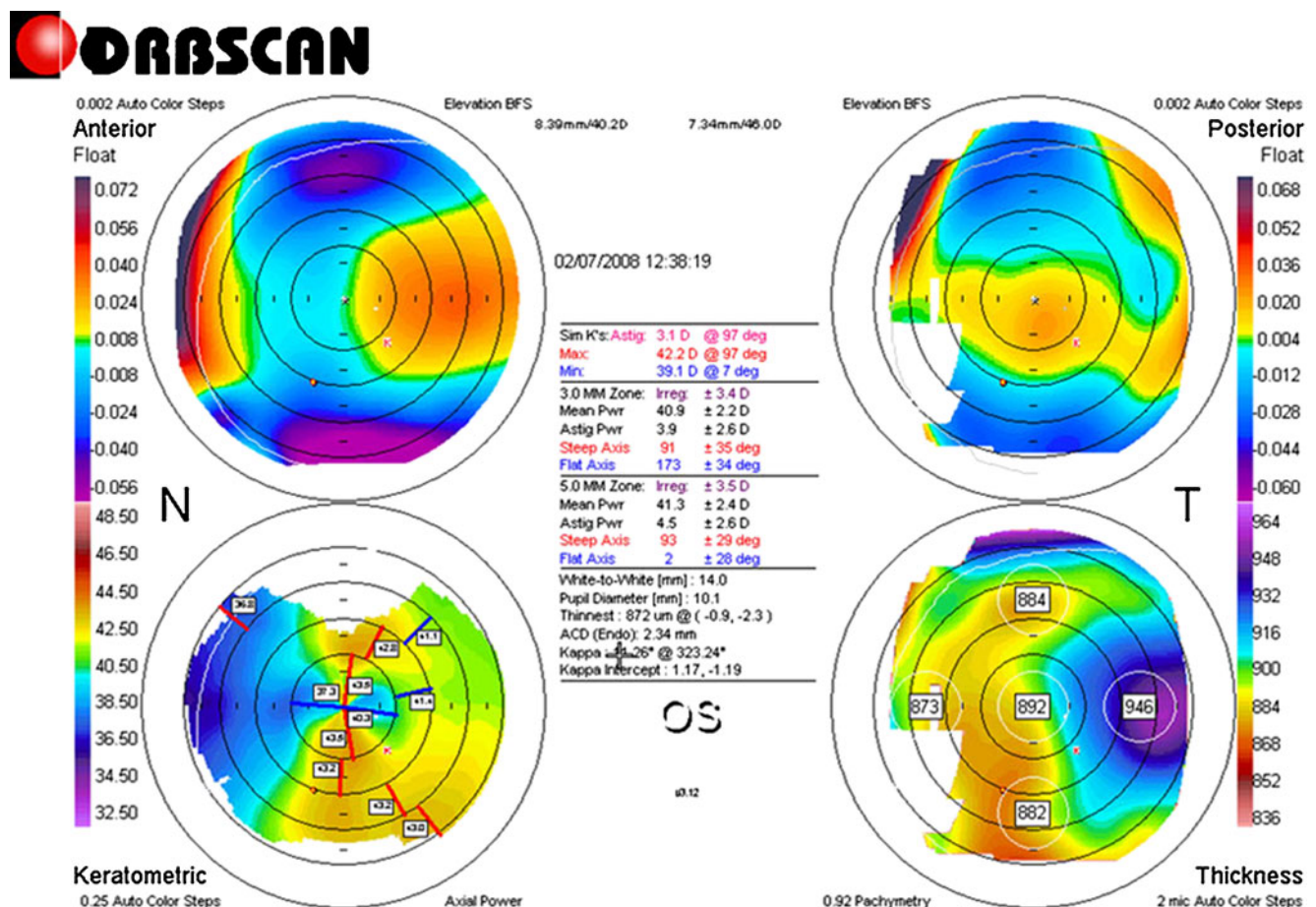


Fig. 1 Slit-scan topography of the pig cornea. *Top*: anterior corneal curvature (right) and posterior corneal curvature (left) represented with respect to best-fit sphere calculated by the topography. *Below*: simulated keratometry (right) and corneal pachymetry (left)

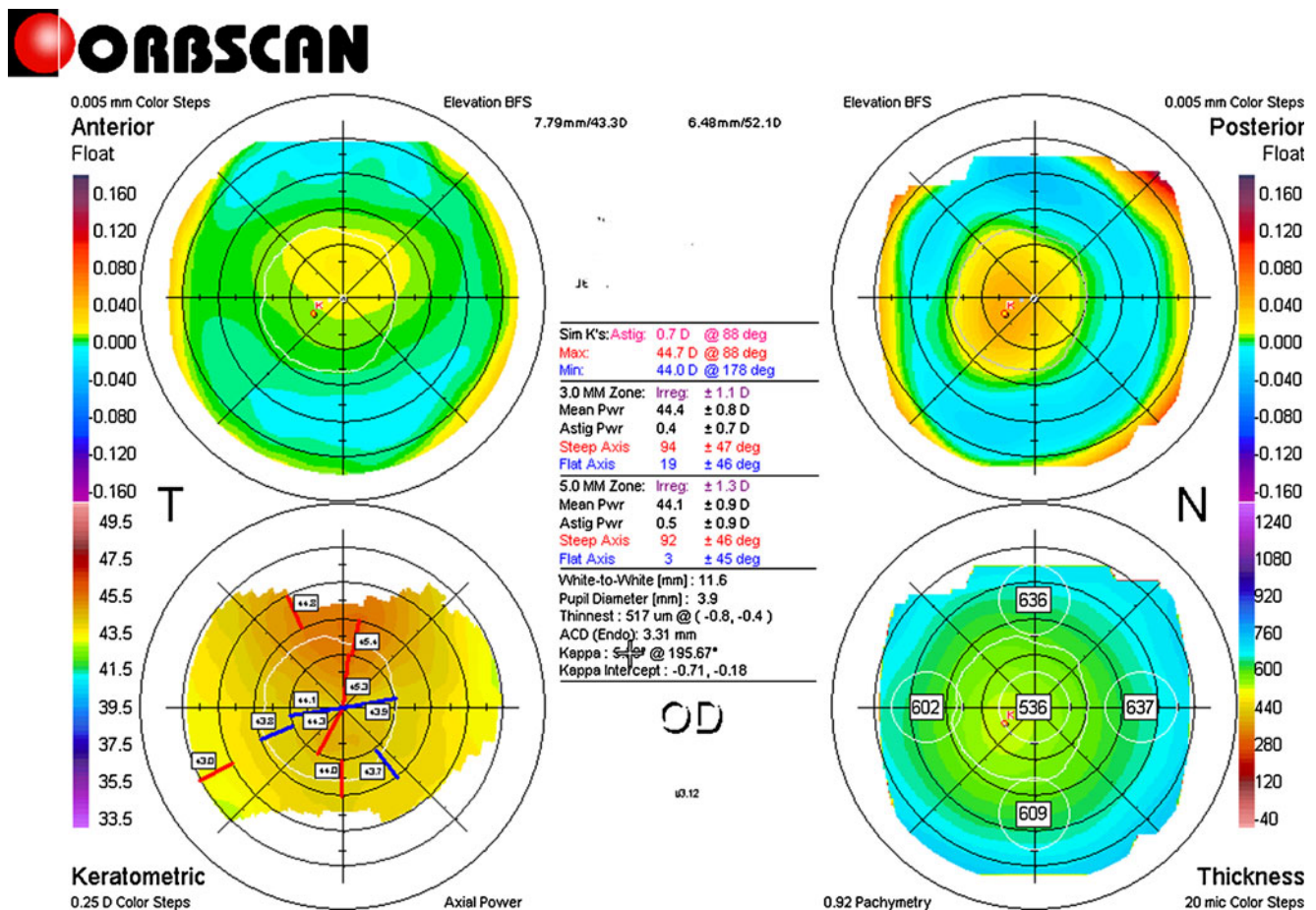


Fig. 2 Slit-scan topography of the human cornea. *Top:* Anterior corneal curvature (*right*) and posterior corneal curvature (*left*) represented with respect to best-fit sphere calculated by the topography. *Below:* simulated keratometry (*right*) and corneal pachymetry (*left*)

curvature, which is easily measurable with actual topography instruments or automatic keratometry. However, our results should be used with caution, because topographical measurements obtained in enucleated eyes and in live studies could be necessary. Pig corneal diameter is larger [18], with a larger corneal radius [9] and greater corneal astigmatism than human cornea. These topographical differences, coupled with corneal thickness differences, would discourage the use of pig eyeball as a model of corneal refractive surgery.

Lens

Some pig lens proteins (α -crystallin and β -crystallin) show the same protein sequence found in human crystallin proteins [21].

The anterior radius of the human lens is 10 mm, and the posterior radius is 6 mm, with a central lens thickness of 4 mm. These radii are smaller than those of the porcine eye. Also, central thickness is less than that of the pig eye [3]. For this reason, the use of the pig eye as an animal model in the development of intraocular lenses is difficult [5].

Ciliary body

The pig ciliary muscle is smaller than in the human eye, and has a circumferential area in the temporal quadrant, which appears to be responsible for accommodation [14].

The study of Wagner et al. [24] using enucleated porcine eyes showed that uveoscleral drainage contributes to the total drainage of the aqueous humour and that the choroid does not represent a significant pathway for uveoscleral drainage.

Retina and vitreous

The porcine retina shows great similarity to the human retina, except for its holangiotic vascularisation and macular organisation [25]. The macula has a band form, but photoreceptor cell density is comparable with that of the human macula [16, 25, 28, 29, 31].

The pig central vitreous mechanical properties are very similar to the human vitreous. [34] These similarities allow the pig eye to be used as a model for retinal pathologies. [1]

Extrinsic ocular motility

The seventh extraocular muscle, the retractor bulbi, must especially be taken into account in ocular surgery experimental studies, because the retraction of the eyeball into the orbit may complicate the surgery [25].

Eyelids

The nictitating membrane structure has degenerated to the caruncle and semilunar fold [25].

Axial length

The axial length found in this study is greater than the one obtained by Bartholomew [8]. The difference may be due to the age of the animal, but Bartholomew did not describe the exact age of experimental animals, only using descriptions such as “young adult” [8], whereas our study used 6- to 8-month-old pigs.

Inflammatory reaction of pig eyeball

The pig eye is more reactive than the human eye [7, 25], indicating that further investigation is needed to determine the cause. Warfvinge [40] showed that the immune privilege of the porcine eye allows a lower inflammatory reaction than that of the human eye. It also might be interesting to know why, in general, the pig eye structures seem to be more pigmented than the human eye, and to know the impact of this pigmentation on the use of the pig eye as an animal model in the visual sciences [16].

Conclusions

This paper describes the anatomy of the pig eyeball collected from the findings described in the literature and from our corneal study. It also summarises the differences between human and pig eye for easy use and interpretation by researchers who are choosing an animal model. The main limitations of our experimental study are the limited number of eyes investigated and the fact that the measurements were conducted on dead animals. Therefore, further measurements in other age groups of the same species will benefit the value of the measurements presented.

Acknowledgments The authors thank all slaughterhouse staff at Justino Gutiérrez SL (Laguna de Duero, Valladolid, Spain) for the cooperation in providing samples used in this work. I. Fernandez-Bueno is supported by the “Junta de Castilla y León”.

References

1. Fernandez-Bueno I, Pastor JC, Gayoso MJ, Alcalde I, Garcia MT (2008) Müller and macrophage-like cell interactions in an organotypic culture of porcine neuroretina. *Mol Vis* 14:2148–2156
2. Acosta E, Vázquez D, Castillo LR (2009) Analysis of the optical properties of crystalline lenses by point-diffraction interferometry. *Ophthalmic Physiol Opt* 29:235–246
3. Wong KH, Koopmans SA, Terwee T, Kooijman AC (2007) Changes in spherical aberration after lens refilling with a silicone oil. *Invest Ophthalmol Vis Sci* 48:1261–1267
4. Ruiz-Ederra J, García M, Hernández M, Urcola H, Hernández-Barbáchano E, Araiz J, Vecino E (2005) The pig eye as a novel model of glaucoma. *Exp Eye Res* 81:561–569
5. Nishi O, Nishi K, Nishi Y, Chang S (2008) Capsular bag refilling using a new accommodating intraocular lens. *J Cataract Refract Surg* 34:302–309
6. Kim MK, Oh JY, Ko JH, Lee HJ, Jung JH, Wee WR, Lee JH, Park CG, Kim SJ, Ahn C, Kim SJ, Hwang SY (2009) DNA microarray-based gene expression profiling in porcine keratocytes and corneal endothelial cells and comparative analysis associated with xeno-related rejection. *J Korean Med Sci* 24:189–196
7. Faber C, Wang M, Scherfig E, Sørensen KE, Prause JU, Ehlers N, Nissen MH (2009) Orthotopic porcine corneal xenotransplantation using a human graft. *Acta Ophthalmol* 87:917–919
8. Bartholomew LR, Pang DX, Sam DA, Cavender JC (1997) Ultrasound biomicroscopy of globes from young adult pigs. *Am J Vet Res* 58:942–948
9. Newell FW (1993) *Oftalmología fundamentos y conceptos*, 7th edn. Mosby, España
10. Forrester J, Dick A, McMenamin P, Lee W (1996) *The eye basics sciences in practice*. Saunders, London
11. Pesudovs K (2004) Autorefractometry as an outcome measure of laser in situ keratomileusis. *J Cataract Refract Surg* 30:1921–1928
12. Nicoli S, Ferrari G, Quarta M, Macaluso C, Santi P (2009) In vitro transscleral iontophoresis of high molecular weight neutral compounds. *Eur J Pharm Sci* 36:486–492
13. McMenamin PG, Steptoe RJ (1991) Normal anatomy of the aqueous humour outflow system in the domestic pig eye. *J Anat* 178:65–77
14. May CA, Skorski LM, Lütjen-Drecoll E (2005) Innervation of the porcine ciliary muscle and outflow region. *J Anat* 206:231–236
15. Nicoli S, Ferrari G, Quarta M, Macaluso C, Govoni P, Dallatana D, Santi P (2009) Porcine sclera as a model of human sclera for in vitro transport experiments: histology, SEM, and comparative permeability. *Mol Vis* 15:259–266
16. Voss Kyhn M, Kiilgaard JF, Lopez AG, Scherfig E, Prause JU, la Cour M (2007) The multifocal electroretinogram (mfERG) in the pig. *Acta Ophthalmol Scand* 85:438–444
17. Jay L, Brocas A, Singh K, Kieffer JC, Brunette I, Ozaki T (2008) Determination of porcine corneal layers with high spatial resolution by simultaneous second and third harmonic generation microscopy. *Opt Express* 16:16284–16293
18. Faber C, Scherfig E, Prause JU, Sørensen KE (2008) Corneal thickness in pigs measured by ultrasound pachymetry in vivo. *Scand J Lab Anim Sci* 35:39–43c
19. Elsheikh A, Alhasso D (2009) Mechanical anisotropy of porcine cornea and correlation with stromal microstructure. *Exp Eye Res* 88:1084–1091
20. Xu YG, Xu YS, Huang C, Feng Y, Li Y, Wang W (2008) Development of a rabbit corneal equivalent using an acellular corneal matrix of a porcine substrate. *Mol Vis* 14:2180–2189
21. Keenan J, Orr DF, Pierscionek BK (2008) Patterns of crystallin distribution in porcine eye lenses. *Mol Vis* 14:1245–1253

22. Shiratani T, Shimizu K, Fujisawa K, Uga S, Nagano K, Murakami Y (2008) Crystalline lens changes in porcine eyes with implanted phakic IOL (ICL) with a central hole. *Graefes Arch Clin Exp Ophthalmol* 246:719–728
23. Chong C, Suzuki T, Totsuka K, Morosawa A, Sakai T (2009) Large coherence length swept source for axial length measurement of the eye. *Appl Opt* 48:144–150
24. Wagner JA, Edwards A, Schuman JS (2004) Characterization of uveoscleral outflow in enucleated porcine eyes perfused under constant pressure. *Invest Ophthalmol Vis Sci* 45:3203–3206
25. Bertschinger DR, Beknazar E, Simonutti M, Safran AB, Sahel JA, Rosolen SG, Picaud S, Salzmann J (2008) A review of in vivo animal studies in retinal prosthesis research. *Graefes Arch Clin Exp Ophthalmol* 246:1505–1517
26. Pandav S, Morgan WH, Townsend R, Cringle SJ, Yu DY (2008) Inability of a confocal scanning laser doppler flowmeter to measure choroidal blood flow in the pig eye. *Open Ophthalmol J* 2:146–152
27. Agopov M, Lomb L, La Schiazza O, Bille JF (2009) Second harmonic generation imaging of the pig lamina cribrosa using a scanning laser ophthalmoscope-based microscope. *Lasers Med Sci* 24:787–792
28. Gu P, Harwood LJ, Zhang X, Wylie M, Curry WJ, Cogliati T (2007) Isolation of retinal progenitor and stem cells from the porcine eye. *Mol Vis* 13:1045–1057
29. Beattie JR, Brockbank S, McGarvey JJ, Curry WJ (2007) Raman microscopy of porcine inner retinal layers from the area centralis. *Mol Vis* 13:1106–1113
30. Fang IM, Yang CH, Yang CM, Chen MS (2009) Overexpression of integrin alpha6 and beta4 enhances adhesion and proliferation of human retinal pigment epithelial cells on layers of porcine Bruch's membrane. *Exp Eye Res* 88:12–21
31. Kiilgaard JF, Prause JU, Prause M, Scherfig E, Nissen MH, la Cour M (2007) Subretinal posterior pole injury induces selective proliferation of RPE cells in the periphery in in vivo studies in pigs. *Invest Ophthalmol Vis Sci* 48:355–360
32. Ng YF, Chan HH, Chu PH, To CH, Gilger BC, Petters RM, Wong F (2008) Multifocal electroretinogram in rhodopsin P347L transgenic pigs. *Invest Ophthalmol Vis Sci* 49:2208–2215
33. Annaka M, Okamoto M, Matsuura T, Hara Y, Sasaki S (2007) Dynamic light scattering study of salt effect on phase behavior of pig vitreous body and its microscopic implication. *J Phys Chem B* 111:8411–8418
34. Swindle KE, Hamilton PD, Ravi N (2008) In situ formation of hydrogels as vitreous substitutes: viscoelastic comparison to porcine vitreous. *J Biomed Mater Res A* 87:656–665
35. Peng YW, Hao Y, Petters RM, Wong F (2000) Ectopic synaptogenesis in the mammalian retina caused by rod photoreceptor-specific mutations. *Nat Neurosci* 3:1121–1127
36. Ninomiya H, Inomata T (2006) Microvascular anatomy of the pig eye: scanning electron microscopy of vascular corrosion casts. *J Vet Med Sci* 68:1149–1154
37. Klećkowska-Nawrot J, Dziegiel P (2007) Morphology of the third eyelid and superficial gland of the third eyelid on pig fetuses. *Anat Histol Embryol* 36:428–432
38. Klećkowska-Nawrot J, Dziegiel P (2008) Morphology of deep gland of the third eyelid in pig foetuses. *Anat Histol Embryol* 37:36–40
39. Olsen TW, Sanderson S, Feng X, Hubbard WC (2002) Porcine sclera: thickness and surface area. *Invest Ophthalmol Vis Sci* 43:2529–2532
40. Warfvinge K, Kiilgaard JF, Klassen H, Zamiri P, Scherfig E, Streilein W, Prause JU, Young MJ (2006) Retinal progenitor cell xenografts to the pig retina: immunological reactions. *Cell Transplant* 15:603–612